

NATHAN PEREIRA

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SKILLS

Engineering Softwares: Siemens NX, Solidworks, Fusion 360, AutoCad 2D/3D, Blender, Revit, CNC Machining, Ansys

Portfolio: <https://njpereira.com>

EDUCATION

University Of Waterloo

Bachelor Of Applied Science, Mechanical Engineering

Waterloo, Ontario

September 2021 - April 2026

WORK EXPERIENCE

Dentra | *Co-Founder*

December 2024 - September 2025

- Achieved award by Conrad business school, receiving two paid interns for summer 2025, a total of **\$10,000** investment, in addition winning **University of Waterloo's \$500 Cornerstone award**.
- Successfully developed **Quantitative light fluorescence** and a rooted deep learning model to detect levels of plaque build up, being the first of its kind to be adaptable to everyday phones. Incorporating manufacturing methods of **silicone-based mold design** and **pcb design**.

Solid Ultrabattery Inc | *Battery Development Intern*

September 2024 - December 2024

- Researching and testing graphite anodes for cells, utilizing **thermogravimetric analysis** and **X-ray fluorescence testing** to determine material composition.
- Designed and tested a peristaltic pump and ventilation system for optimizing the cathode coating process, achieving **50%** reduction in production time, utilizing **SolidWorks** for design and **Ansys** for **CFD analysis**.
- Developed 3D-printed single and multi-layer pouch cell molds, streamlining the manufacturing process and achieving a **25%** reduction in production time.

Tesla | *R&D Process Engineering Intern - Cell Engineering*

January 2024 - April 2024

- Led the testing and validation of POP designs to improve **40 electrolyte filling machines** in Texas Gigafactory, utilizing in-depth procedures and root cause analysis to optimize designs.
- Spearheaded soak testing within a lab environment to evaluate the electrolyte compatibility of **18 material compositions** within cell manufacturing, applying a combination of qualitative and quantitative analysis.
- Conducted research and spearheaded the development of an RFID system slated for integration into upcoming next-generation electrolyte machines, representing an investment of over **\$2.3 million**. Responsibilities encompassed rigorous wet lab material testing, design work, and validation of proof of concept.
- Conducted thorough **fatigue and dispense calibration cycling tests** on electrolyte pumps to pinpoint root causes and provide valuable feedback to designers. Utilized a **VHX microscope** for detailed analysis of degradation over time, enhancing our understanding of long-term performance.

Toyota | *Analyst Engineering Intern*

May 2023 - August 2023

- Utilized Toyotas' TBP problem analysis method to lead a cross-functional team in the complete redesign of the magnetic path for AGV navigation, optimizing route planning and eliminating potential disruptions, leading to improved overall operational reliability and a **62% decrease** in AGV errors.
- Engineered and implemented 3D-designed ergonomic hood jigs and fixtures, improving production efficiency and eliminating mechanical failures, resulting in **\$5,000** in cost savings.

Cemcorp | *Mechanical Engineering Intern*

September 2022 - December 2022

- Prepared comprehensive process flow charts, **P&ID diagrams**, BOM, tie-point list, and equipment lists to aid seamless communication with stakeholders throughout the design and implementation phases of an industrial alcohol distillery plant.
- Designed 3D piping models for **stress analysis** and developed a comprehensive drawing package for an alcohol distillery plant, ensuring **ISO 9001 compliance**, which led to a **Co-op Student of the Year nomination**.

MCW Consultants | *Mechanical Engineering Intern*

January 2022 - April 2022

- Analyzed mechanical blueprints and markups, providing drafting of HVAC and Plumbing in **AutoCad** and **Revit** 3-D modelling for 10 different clients

PROJECTS/DESIGN TEAMS

Battery Workforce Challenge - Solidworks, Ansys

Designed an **EV battery module** using Solidworks, conducting research on structural materials, performing load simulations and ensuring all mechanical and structural standards are met for the competition.

UWMedTechResolve - Solidworks, Project Management

One of the team leads for the **conceptual design and development phase** of the University of Waterloos Medtech design team for the upcoming Form and Function competition. Performing **design sprints**, **R&D** and ensuring deliverables are met while leading **15 members**.